Packet work and live-storage recurrence: review proposal

Source review prepared 7 November; registered 8 November 2026 as PCC-20261108-ebd943, kind statement, standing UNVERIFIED. This is a source-pinned budget specification for review, not a proved bound, executed control or verified PCC result. It has no proof or signed verdict. The owner authorized one official ledger append, preserving every prior row. The remaining integration boundary is recorded in REGISTRATION.md (REGISTRATION.md). All proposed arguments below remain unverified. [pcc:unverified]

The proposed advance joins packet preparation, every attempted DFS choice, outgoing integer arithmetic, child construction, interpolation and retained ancestors in one cost ledger. The coarse entry budget uses the full current mode count B; it does not require the unverified q<=E<=R candidate. That candidate remains relevant to tuple/child correctness and sharper budgets. This proposal supplies conditional inequalities and a trace-accounting recurrence, not a checked complexity theorem. [pcc:unverified]

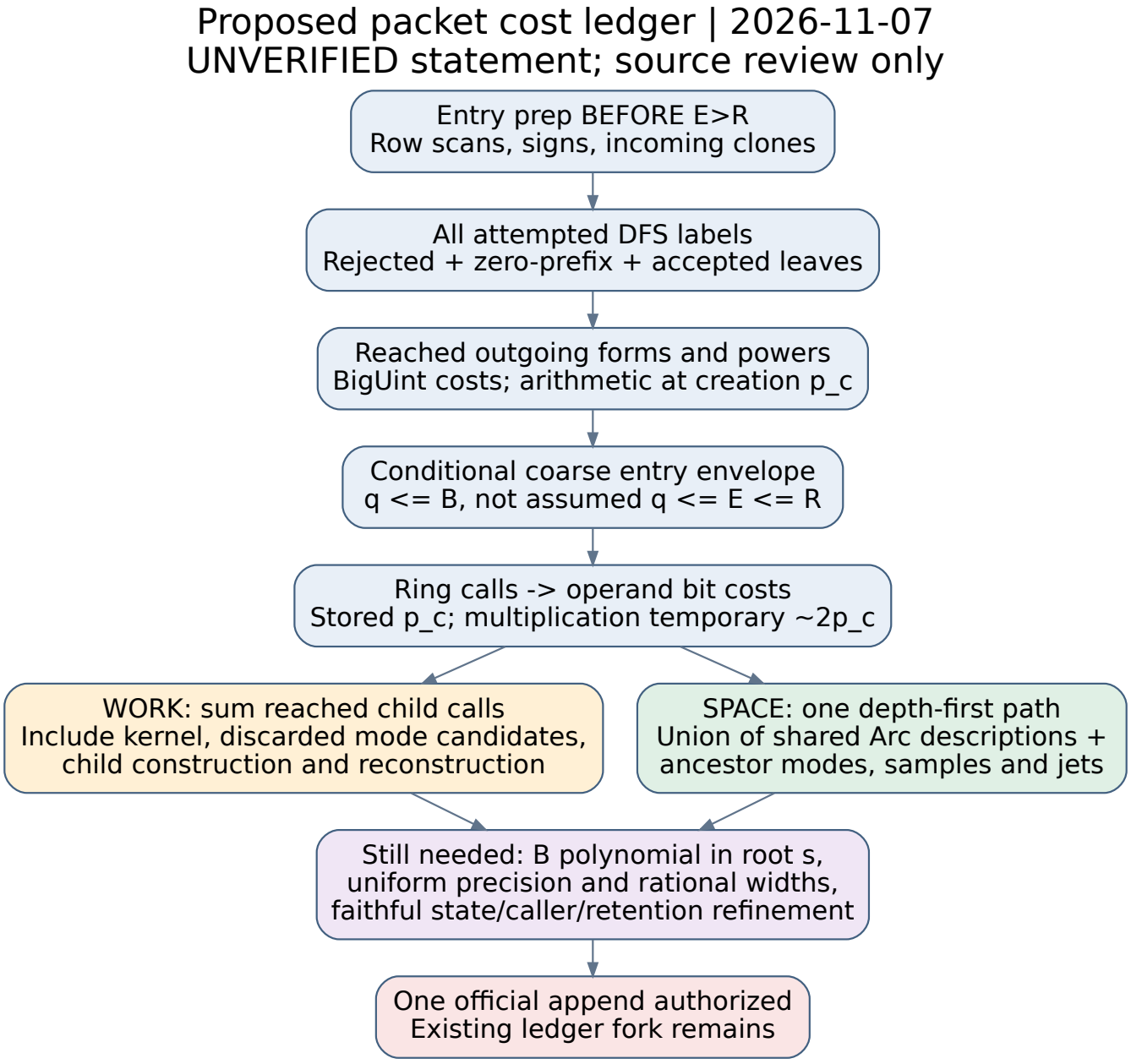


Figure 1: Evidence dependencies; no timing data.

Source scope and obligations

Source pin: crypto 08fb3510592ce4849abaf660bbee88191d3d6263, the published head of PR3569. The native source still has the Nov6 audited bodies; current main metadata and prior packets are separately pinned in sources.json. No whole-program parity is inferred. The prior DFS argument and Nov5 tuple argument are unverified inputs, not checked lemmas. This study leaves every old packet, claim and event intact. [pcc:unverified]

Primary source: OpenAI Appendix D.1/D.4.5-D.4.7, PDF last-path change 42907c9e999cc00d6abe7a50bfc1023250edd1b9, blob 0503566e95bc587a4076573a45132d97531b8ba6. D.1 explicitly permits exponential entry work and polynomial entry space. D.4.6 prices expansion of at most L row factors and repeated squaring; D.4.7 accumulates depth-first state and Taylor data. This is the existing audit target, not a new identity claim. [pcc:unverified]

Sources checked (KB): tools absent, wrapper status exits127 (no executable). The intended queries and unavailable retrieval are in KB-AND-ATLAS.txt. Refreshed atlas dfc2d3793c5693404f1147c6e1c954f1e0eafd21 still assigns this to LEAD-EXO-hashed-factor-base-tensor-ecdl's independent audit. Retrieval scores are not evidence; no full-delta stamp or novelty conclusion.

Dimensions and preconditions

Symbol	Definition; not a measured value
s, L_desc	Root edge scale and total original graph/tensor-description length
m, ell, K, B	Current edges, packet slot capacity, packets, retained modes
R,C,q,h	One stored packet's rows, columns, incoming and outgoing records
beta_j,E	Incoming target exponents and their sum
d,p_c,p	Ring coordinates, immutable packet creation precision, requested precision
a_c	Number of selected column bits for this entry
n_false	Number of false row bits
A,P,F	Attempted candidate labels, committed pushes, eligible visited leaves
h_run,S_run	Outgoing records reached and their selected sparse terms scanned
W	Machine word bits; vector metadata and indices are charged separately

Assume valid shapes, canonical coefficients, successful allocations, exact ring operations, nonnegative exponents and representable R+C,R+1,q+1,E. Refining the Ok-return path is separate from charging an Err prefix. A limit or arithmetic error is not an answer or complete traversal. Original Boolean callbacks need D.1's own uniform evaluation hypothesis; their trait type supplies no bound. [pcc:unverified]

One packet entry: a conditional work envelope

Define Add(d,p_c), Sub(d,p_c), Mul(d,p_c), Clone(d,p_c), Zero(d,p_c), and Lift(d,p_c,p) as bit costs, including canonicality checks and allocations where applicable. BigAddMod(p_c) prices the outgoing BigUint addition/remainder; its cost is not a Ring::add charge. Define Elem(d,p_c) as an allocated element footprint; a payload proxy d*(p_c+W) is conditional on a capacity bound, not an RSS observation. [pcc:unverified]

Preparation is paid even on the zero-degree exit

Preparing row constants uses R*a_c ring additions and n_false*(q+1) subtractions. True rows clone their q incoming coefficients instead of negating them; container initialization and zero/one construction also cost work. A proposed safe preparation budget is

Prep_entry = R*a_c*Add + n_false*(q+1)*Sub + O(R*q*Clone + (R+q)*Elem + R*C + q).

The R*C term pays bit-index scans even when no selected column causes a ring addition. Exponent-vector creation and summation cost O(q) word work under the sum-width premise. The source performs all this BEFORE testing E>R. If that guard returns zero, the DFS and outgoing costs vanish, not preparation. The expression is an unverified envelope, not an exact meter total. [pcc:unverified]

Traverse attempted labels, not surviving coefficients

Every attempt increments choices[row] and charges the extraction meter before rejecting an exhausted exponent, zero factor or excessive remaining degree. Every committed push calls Ring::mul; every eligible leaf calls Ring::add. The source does not prune zero prefix products: zero divisors do not allow us to remove those visits from A, P or F. Signs and indexed multiplicities stay as in the previous source-semantic target. [pcc:unverified]

For review, let b=q+1. A coarse full-tree envelope is

A <= b * sum_{r=0..R-1} b^r; P <= sum_{r=1..R} b^r; F <= b^R.

Empty sums are zero; when q=0, use A<=R,P<=R,F<=1 without division by b-1. These are candidate bounds obtained by embedding the cursor traversal in the ordered row-choice tree; a faithful state/termination refinement is still required. Pruning only lowers visits if that refinement holds. The earlier unverified restoration/capacity argument is not being silently promoted to a checked lemma. [pcc:unverified]

Proposed DFS bit work:

DFS_entry <= O((A+P+F+R+q)*W + A*Zero(d,p_c)) + P*Mul(d,p_c) + F*Add(d,p_c).

This deliberately includes rejected labels and all backtracking. It may charge zero tests for attempts that skip before a factor is obtained; that is a safe overcharge under the stated model, not a measurement. [pcc:unverified]

Outgoing arithmetic and final lift

For each reached outgoing record with exponent e, Ring::pow calls mul exactly popcount(e) + max(bitlength(e)-1,0) times on a successful path. It then multiplies the coefficient sum by that power once. A proposed outgoing budget is

Out_entry <= S_run*BigAddMod(p_c) + sum_reached [Embed + (popcount(e)+max(bitlength(e)-1,0)+1)*Mul] + scan/zero/allocation work.

Count all sparse-term bit checks, including unselected terms; S_run in the arithmetic term counts selected terms, while the separate scan term counts all reached terms. A zero initial coefficient skips every outgoing record; a later zero product skips only the remaining records. e=0 still constructs and multiplies one; child normally omits such records. No factorial or mode deduplication is introduced. [pcc:unverified]

Packet arithmetic uses p_c throughout. Tensor::entry validates requested precision and creation identity, then lifts the completed packet result. Thus Entry_work <= Prep_entry + DFS_entry + Out_entry + Lift + overhead. Substituting p for p_c inside earlier arithmetic would change semantics and would also overstate which widths actually occurred. [pcc:unverified]

Translating ring calls into bit work

Ring::add/sub/check scan d canonical coordinates. Ring::mul has at most d^2 coordinate products and O(d^2) reduction updates, with immediate masking. Canonical input coordinates are below 2^p_c, so a product/update temporary can need about 2*p_c+1 bits even though stored outputs use p_c bits. A conditional school-operation envelope is

Mul(d,p_c) <= O(d^2 * (M_int(p_c)+p_c) + d*p_c + allocation work).

M_int is the actual arbitrary-precision multiplication cost, not one unit. Scratch holds O(d) output coordinates, not a retained d-by-d product table. Ring::pow uses that cost repeatedly; BigUint::modpow in mode preparation and BigUint remainder in outgoing forms require their own bit-cost contracts. Ring/unit inverse and initial irreducible-polynomial selection remain explicit cost terms. No implementation complexity or allocator bound is proved here. [pcc:unverified]

Preparation outside the entry must not disappear

Child construction scans all old vertices and allocates port maps, scans every packet's slots, builds its R*C local kernel table, scans ALL B modes per packet, and copies selected incoming/outgoing coefficient records. Incoming lookup is linear in mode.rows for each selected row; outgoing lookup is linear in mode.columns for each selected column. Cloning an Arc is cheap reference work; cloning an Element or BigUint copies coefficient storage. [pcc:unverified]

A proposed construction ledger is

ChildPrep = graph/map/slot/edge work + sum_packets [R*C*KernelWeight + B*ModeTest + sum_in R*(RowLookup+Clone) + sum_out C*(ColumnLookup+Clone)].

KernelWeight is a creation-ring operation cost, not an entry callback. New packets contain coefficient data and no parent evaluator. They do not retain the whole current mode vector, but the parent evaluate_at frame keeps that vector live while the child is evaluated. [pcc:unverified]

Mode construction tries packet/tile/weight candidates before discarding empty or zero records. Charge their column modpow calls and row mode_coefficient calls, including discarded candidates. With tile count 2^(i+1), a crude candidate bound is K*2^(i+1)*max(p-1,0), with separate slot/row scans and per-candidate column/row work. Stored B alone is not the preparation counter. Cauchy::new builds its grid/factor inverses before tuple enumeration. All these are definitions of terms to bound, not established global polynomial bounds. [pcc:unverified]

Whole-call work recurrence

Let T(n) price one successful contract call n, including its descendants. For an active node, I=m*(2*ell-1)+1 samples are used at every active t; local precision p_t follows child_precision. Let tuples(n,t,j) denote the tuples actually visited, and groups(beta) the children reached before a zero-product shortcut. A checkable trace-accounting specification is

T(n) = FrameSetup + Pack(n) + sum_t [RingView(t) + SampleSetup + sum_{j<I} (At(n,t,j)+SampleStore) + Interpolate(I,d,p_t) + JetStore(t)] + Recover(n) + Cleanup

At = Kernel + Modes_attempted + BoundsInit + TupleSetup + sum_beta [ProductInit + sum_{g in groups(beta)} (ChildPrep_g + T(child_g) + Mul(d,p_t) + ZeroTest + GroupControl) + Add(d,p_t) + NextTupleScan] + Cleanup.

Initial ring construction/irreducible selection and Setup are a separate root setup term, not hidden inside contract. The final NextTupleScan that returns false is included. Empty modes still produce the initial zero tuple and one group loop. An Err is charged only through its executed prefix, not treated as a successful term. Neither recurrence is a newly certified mathematical equality; it is the proposed partition of source work that a refinement/operation model must check. [pcc:unverified]

At a fallback leaf with m edges, retain every visited assignment and vertex entry before the product-zero break. A conservative envelope is

T_leaf <= 2^m * [assignment/port work + sum_vertices (Entry_work(v) + Mul(d,p)) + Add(d,p)].

Original predicates use their supplied bound; retained packet descriptions use their own p_c and local dimensions. Do not replace every entry by the cheapest newly created packet. Public machine guards/fallback remain unchanged. [pcc:unverified]

Live storage: one recursive path plus reachable descriptions

A new packet's stored logical payload includes

Desc(P) = R*C*Elem(d,p_c) + R*q*Elem(d,p_c) + outgoing sparse coefficients/indices + exponent/vector/ring metadata.

Entry scratch adds approximately (R*q+2*R+O(1))*Elem(d,p_c) for coefficients, constants and prefix cells, plus O((R+q)*W) counters; outgoing pow adds only a constant number of Elements at a time. Actual Vec/BigUint capacities must be bounded separately. R=0 retains its empty-product behavior. [pcc:unverified]

Let U_path be the union of tensor-description allocations reachable from ALL live ancestor/child graph Arcs. Count each allocation once by identity. A retained Arc shares the same description; a new packet adds a fresh allocation. The active frames also retain graph/packing/port arrays, Cauchy factors, full mode tables, beta/usage/bounds, accumulated products/sums and temporary child construction. [pcc:unverified]

While evaluating the jth sample, nodes/values for previous samples and all previous jets of that frame remain live. After interpolation, current sample arrays coexist with coefficients until scope exit. The Lagrange implementation stores O(I) Elements at once but does polynomial arithmetic repeatedly; recovery retains all jets plus series/quotient scratch and rational numerators/denominators. Charge actual maximum rational bit widths, not p_t alone. Earlier samples/jets from EVERY ancestor remain live during descent. [pcc:unverified]

A proposed peak-storage recurrence is

Peak(path) <= Bytes(U_path) + sum_live_frames [GraphPacking + KernelModes + TupleState + SamplesJets + FrameScalars] + max(CurrentChildPrep, EntryScratch, InterpolationScratch, RecoveryScratch).

Equivalently, a local frame contributes its nonshared live storage plus the maximum reached child peak, with shared descriptions charged by the union. Do NOT sum sibling peaks or multiply the parent entry evaluator into a new packet; do NOT omit ancestors. Samples need up to 2*I Elements; retained jets need sum_previous q_t Elements at their individual p_t. [pcc:unverified]

What is needed to compare with Appendix D

Proposed transfer	Preconditions still to discharge
New-packet entry envelope exp(O(ell*log s)) times polynomial factors	R<=ell, q<=B with B polynomial in s alone, bounded d,p_c and coefficient/allocator costs
D.1 exp(O(s^(9/10))) entry allowance	ell<=s^(1/1000) uniformly; constants fixed; root predicates satisfy original-description hypothesis
Polynomial description/live space	Polynomial K,B,I,total jets, coefficient and rational widths per path; logarithmic-depth/shared-identity retention induction
D.4.7 total call bound	Correct weighted-tuple enumeration/count, child shrinkage/ precision bounds, leaf cutoff and every per-term construction/ recovery cost

The coarse entry envelope needs q<=B, not the sharper q<=E<=R. In particular, B must be polynomial in s itself for the paper's log s exponent: merely polynomial in s+L_desc could put log L_desc inside the exponential and lose D.1's fixed polynomial dependence on original description length. This is a conditional dependency distinction, not an established bound for the native backend. [pcc:unverified]

Disposition: the source recurrence is prepared, but the complete D.1/D.4.7 allowance is NOT discharged. Faithful tuple/ DFS refinement, uniform mode and precision bounds, ring/BigUint/rational bit complexity and the allocation-union induction remain obligations. No finite control was run as a substitute. [pcc:unverified]

Next discriminating step: bind B and all attempted mode-preparation candidates to root s and the tapered p_t along one recursive path, together with exact sample/jet lifetimes. That would close the missing dimensional inputs to this ledger before a faithful machine/refinement proof can claim its bounds.

The general audit remains open. No canonical leaderboard, curve or performance graph obtains new measurements or transfer edges. Existing measured graphs and historical reports stay unchanged. The packet has editable source/PDF and metadata checks. Registration is complete as an unverified statement; PR integration still requires current-head gates, actual protection/reviews and an immutable-preserving fork resolution.